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Homework-5 (Classical and quantum scattering)

1. Consider a classical scattering process by a 3D hard sphere of radius R as shown in the diagram [Fig. 1]. The hard sphere potential is given by $V(r) = \infty$ for $r \leq R$, otherwise zero.

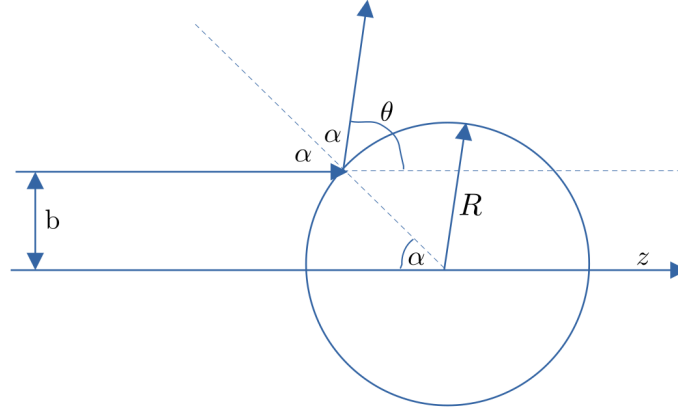


Figure 1: Sketch of a scattering event from a hard sphere scatterer.

- (a) Obtain the relation between the impact parameter b and the scattering angle θ as shown in the diagram.
 - (b) Calculate the differential scattering cross-section $D(\theta) \equiv \frac{d\sigma}{d\Omega}$. Here $d\sigma = b db d\phi$ and $d\Omega = \sin\theta d\theta d\phi$. Note that $\frac{d\sigma}{d\Omega}$ is not the usual derivative!
 - (c) Calculate the total scattering cross-section $\sigma_T = \int D(\theta) d\Omega$.
2. Consider a classical scattering process by a 2D hard disk of radius R .
 - (a) Draw a sketch of a scattering event from a hard disk scatterer. Obtain the relation between the impact parameter b and the scattering angle ϕ . Here the incident particles are moving along x axis and the scattering angle ϕ is measured with respect to the x -axis. Also, $0 \leq \phi \leq 2\pi$.
 - (b) Show that the “differential scattering cross-section” is

$$D(\phi) \equiv \frac{db}{d\phi} = \frac{R}{2} \left| \sin \frac{\phi}{2} \right|.$$

Note the dimension of the differential scattering cross-section.

- (c) Show that the total scattering cross-section is $\sigma_T = \int D(\phi) d\phi = 2R$.
3. **Rutherford scattering:** Consider an incident particle of charge q_1 with kinetic energy E scatters a heavy particle of charge q_2 . Here the scattering potential is the Coulomb potential between the two charges.

- (a) Show that the impact parameter b can be obtained as

$$b = \frac{q_1 q_2}{8\pi\epsilon_0 E} \cot(\theta/2),$$

where θ is the scattering angle.

- (b) Show that the differential scattering cross-section is

$$D(\theta) = \left[\frac{q_1 q_2}{16\pi\epsilon_0 E \sin^2(\theta/2)} \right]^2.$$

- (c) You will see that the total cross-section is not finite. This is because the Coulomb potential has infinite range.

[It is not easy to solve this problem. See the classical mechanics book by Goldstein or any other standard books].

4. **Partial wave analysis:** Consider a particle with energy $E = \hbar^2 k^2 / (2m)$ is moving towards a static hard-sphere of radius R . Our aim is to calculate scattering cross-section quantum mechanically.

- (a) The most general solution outside the hard-sphere potential ($r > R$) can be written as

$$\psi(\mathbf{r}) = \sum_{l=0}^{\infty} i^l (2l+1) P_l(\cos \theta) [a_l j_l(kr) + b_l n_l(kr)],$$

As $r \rightarrow \infty$, the asymptotic expressions of $j_l(x)$ and $n_l(x)$:

$$j_l(x) \sim \frac{\sin(x - l\pi/2)}{x}, \quad n_l(x) \sim -\frac{\cos(x - l\pi/2)}{x}.$$

Use these asymptotic expressions, show that $\psi(\mathbf{r})$ in the asymptotic limit can be expressed as

$$\psi(\mathbf{r}) \sim \sum_{l=0}^{\infty} \frac{(2l+1)}{2i} P_l(\cos \theta) \left[(a_l - ib_l) \frac{e^{ikr}}{kr} + (-1)^{l+1} (a_l + ib_l) \frac{e^{-ikr}}{kr} \right].$$

- (b) The scattering amplitude $f(\theta)$, which is the coefficient of the outgoing wave ($\frac{e^{ikr}}{kr}$) must be a purely imaginary, so we can write $a_l - ib_l = e^{2i\delta_l}$ and $a_l + ib_l = 1$. Show that

$$a_l = \cos \delta_l e^{i\delta_l}, \quad b_l = -\sin \delta_l e^{i\delta_l}.$$

these expansion coefficients are obtained using the asymptotic limits of j_l and n_l . But they will be equally valid for all range of $r > R$. Moreover, we will see that $l = 0$ is the most dominating term and the asymptotic expressions for $l = 0$ match exactly with the exact expressions. Thus we can safely use the expressions of a_l and b_l . Also, $\psi(r)$ we are considering for $r > R$, for large R asymptotic expressions are matching quite well.

Therefore the most general solution outside the hard-sphere potential ($r > R$) can be re-written as

$$\psi(\mathbf{r}) = \sum_{l=0}^{\infty} i^l (2l+1) P_l(\cos \theta) e^{i\delta_l} [\cos \delta_l j_l(kr) - \sin \delta_l n_l(kr)],$$

- (c) Using the boundary condition, $\psi(r = R) = 0$, show that

$$\tan \delta_l = \frac{j_l(kR)}{n_l(kR)}.$$

- (d) Calculate the phase shifts δ_l for the first three partial waves ($l = 0, 1, 2$) for (i) $kR = \pi/6$ and (ii) $kR = \pi/4$.
- (e) **Low-energy approximation:** In the low-energy limit, $kR \ll 1$, we know

$$j_l(kR) \sim \frac{(kR)^l}{(2l+1)!!}, \quad n_l(kR) \sim -\frac{(2l-1)!!}{(kR)^{l+1}}.$$

Show that in the low-energy limit ($kR \ll 1$) the phase shift is

$$\delta_l \sim -\frac{(kR)^{2l+1}}{(2l+1)!!(2l-1)!!}.$$

It clearly shows that the maximum phase shift occurs for $l = 0$ and it decreases with increase of l . The maximum phase shift is $\delta_0 = -kR$.

As l increases, the wavefunction sprades away from the scatterer and so the overlap with the potential decreases. This lead to a smaller phase shift.

- (f) We know that the total cross-section is

$$\sigma_T = \frac{4\pi}{k^2} \sum_l (2l+1) \sin^2 \delta_l.$$

Calculate σ_T keeping contributions from the first three partial waves for (i) $kR = \pi/6$ and $kR = \pi/4$.

- (g) Using the trigonometric relation, $\sin^2 \delta_l = \tan^2 \delta_l / (1 + \tan^2 \delta_l)$, show that the total cross-section in the low-energy limit is

$$\sigma_T \sim 4\pi R^2 [1 + \mathcal{O}(kR)^4 + \dots].$$

This is four times larger than the classical result.

5. Consider a scattering process by a spherically symmetric potential well

$$\begin{aligned} V(r) &= -V_0, \quad r \leq R \\ &= 0, \quad r > R. \end{aligned}$$

Show that in the low-energy limit $kR \ll 1$, the total cross-section for $l = 0$ partial wave is given by

$$\sigma_T = 4\pi R^2 \left[1 - \frac{\tan(qR)}{qR} \right]^2,$$

where $q^2 = 2m(E + V_0)/\hbar^2 = k^2 + 2mV_0/\hbar^2$ with m being mass of the incident particle.

6. Consider a scattering process by a three-dimensional δ -function shell of radius R as described by $V(r) = V_0\delta(r - R)$.

- (a) Show that the s-wave phase shift δ_0 is given by

$$\sin^2 \delta_0 = \frac{\beta^2 \sin^4 x}{x^2 + x\beta \sin(2x) + \beta^2 \sin^2 x}.$$

Here $x = kR$ and $\beta = 2mV_0R/\hbar^2$.

- (b) Show that the total cross-section in the low-energy limit is

$$\sigma_T = 4\pi R^2 \left[\frac{\beta}{1 + \beta} \right]^2.$$

7. Show that

$$G(\mathbf{r}, \mathbf{r}') \equiv \int \frac{d^3 k'}{(2\pi)^3} \left[\frac{e^{i\mathbf{k}' \cdot (\mathbf{r} - \mathbf{r}')}}{k^2 - (k')^2 \pm i\epsilon} \right] = -\frac{1}{4\pi} \frac{e^{\pm ik|\mathbf{r} - \mathbf{r}'|}}{|\mathbf{r} - \mathbf{r}'|}$$

8. Show that

$$\psi(\mathbf{r}) = e^{i\mathbf{k} \cdot \mathbf{r}} + \frac{2m}{\hbar^2} \int d^3 r' \left[-\frac{e^{ik|\mathbf{r} - \mathbf{r}'|}}{4\pi|\mathbf{r} - \mathbf{r}'|} \right] V(\mathbf{r}') \psi(\mathbf{r}')$$

is the solution of the following Schrödinger equation

$$\left[-\frac{\hbar^2}{2m} \nabla_{\mathbf{r}}^2 + V(\mathbf{r}) \right] \psi(\mathbf{r}) = E\psi(\mathbf{r}),$$

with $E = \hbar^2 k^2 / (2m)$.

9. **Born approximation:** The scattering amplitude within the first-order Born approximation is given by

$$f_B(\theta) = -\frac{m}{2\pi\hbar^2} \int d^3 r' e^{i\mathbf{q} \cdot \mathbf{r}'} V(\mathbf{r}').$$

Here m is the mass of the incident particle, $V(\mathbf{r})$ is the scattering potential, $\mathbf{q} = \mathbf{k}_s - \mathbf{k}$ is the change in the wave vector and $q^2 = 2k^2(1 - \cos \theta)$.

- (a) Calculate the scattering amplitude within the Born approximation for the following scattering potential:

$$V(\mathbf{r}) = V_0 \lambda^3 \delta^3(\mathbf{r}), \quad V(\mathbf{r}) = V_0 \lambda^5 \nabla^2 \delta^3(\mathbf{r}), \quad V(\mathbf{r}) = V_0 e^{-r^2/(4\lambda^2)}.$$

Here V_0 has the dimension of energy and λ has the dimension of length.

- (b) **Soft sphere potential:** Consider the following finite spherical potential barrier:

$$\begin{aligned} V(r) &= V_0, & r &\leq R \\ &= 0, & r &> R. \end{aligned}$$

- (i) Show that the scattering amplitude is

$$f_B(\theta) = -\frac{2mV_0R^3}{\hbar^2} \left[\frac{\sin(qR) - (qR) \cos(qR)}{(qR)^3} \right],$$

where $q = 2k \sin(\theta/2)$.

For a given $kR = 1/2$, find the scattering angles θ at which $f_B(\theta) = 0$.

- (ii) Show that the total scattering cross-section is

$$\sigma_T = \pi R^2 \left[\frac{2mV_0 R^2}{\hbar^2} \right]^2 F(d),$$

where the function $F(d)$ is given by

$$F(d) = \left[\frac{2d^4 - 2d^2 - 1 + 2d \sin(2d) + \cos(2d)}{d^6} \right],$$

with $d = 2kR$.

- (iii) Plot $F(d)$ vs d . Show that $F(d) \sim 4/9 - 2d^2/(45) + \dots$ as $d \rightarrow 0$. [Take the limit $d \rightarrow 0$ carefully].

From the plot, show that the maximum value of $F(d)$ is $4/9$ at $d = 0$.

- (iv) Show that the maximum total cross-section is

$$\sigma_T^{\max} = 4\pi R^2 \left[\frac{V_0}{3} \frac{2mR^2}{\hbar^2} \right]^2.$$

10. Using the Lippmann-Schwinger formalism, calculate the transmission and reflection coefficients in 1D scattering problem where the scattering potential is given by

$$V(x) = \frac{\hbar^2}{2m} \gamma \delta(x).$$

Here γ is a constant. The results will match exactly with the results obtained in the conventional method.

11. **Connection between the Fermi's golden rule and the differential cross-section within the first-order Born approximation.** Consider an elastic scattering process where a free particle with momentum $\mathbf{p}_i$ is scattered to the momentum $\mathbf{p}_f$ by the scattering potential $V(\mathbf{r})$.

According to the Fermi's golden rule, the transition rate into the final states entering into the solid angle $d\Omega$ is

$$\Gamma_{fi} = \frac{2\pi}{\hbar} \int_{p=0}^{\infty} |V_{fi}|^2 \delta \left(\frac{\mathbf{p}_f^2}{2m} - \frac{\mathbf{p}_i^2}{2m} \right) \frac{d^3 r p^2 dp d\Omega}{(2\pi\hbar)^3},$$

Here $V_{fi} = \langle f | V(\mathbf{r}) | i \rangle$ is the matrix elements of the scattering potential V between the two free particle states. Note that we are integrating over the radial momentum p only, not on the angular variables.

Show that

$$\begin{aligned} \frac{d\sigma}{d\Omega} &= \frac{1}{j_{\text{in}}} \frac{\Gamma_{fi}}{d\Omega} \\ &= |f_B(\theta)|^2. \end{aligned} \quad (1)$$

where $j_{\text{in}} = p_f/(mL^3)$ with L being the size of the box.